

Freight Rate Prediction Challenge

Machine Learning Engineer Assessment Report

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Role: Machine Learning Engineer

1. Introduction

The objective of this project was to develop a machine learning model capable of accurately predicting freight rates using historical shipment data. The model was trained using the provided development dataset and evaluated before generating predictions for the unseen validation dataset.

The final solution includes data preprocessing, feature engineering, model comparison, prediction generation, and evaluation using the provided scoring script.

2. Dataset

The project utilized the following datasets provided as part of the assessment:

train_test.csv – Labeled dataset used for model training and validation.

validation.csv – Unseen dataset requiring freight rate predictions.

validation_predictions_template.csv – Template used to generate the final submission.

december_chart_inputs.csv – Dataset used for generating the December prediction chart.

3. Data Exploration

The datasets were loaded into Google Colab for analysis. Initial exploration included:

Understanding the structure and data types of each feature.

Checking for missing values.

Reviewing numerical and categorical features.

Exploring the target variable (posted_rate).

This exploratory analysis helped identify the preprocessing steps required before model training.

4. Data Preprocessing

Several preprocessing techniques were applied to improve data quality and prepare the data for machine learning models.

These included:

Handling missing values using median imputation.

Extracting useful date-based features such as month, day, week, quarter, and day of the week.

Frequency encoding for pickup and delivery locations.

One-hot encoding for equipment type.

Aligning feature columns across the training, validation, and December datasets to ensure consistency.

5. Feature Engineering

Feature engineering was performed to improve model performance by creating meaningful predictive variables.

The final feature set included:

Pickup latitude and longitude

Delivery latitude and longitude

Distance

Shipment weight

Market index

Quote signal

Date-related features

Frequency-encoded pickup and delivery locations

Encoded equipment type

6. Training and Validation Approach

The labeled development dataset (train_test.csv) was divided into training and validation sets using an 85% training and 15% validation split.

Multiple regression models were trained and evaluated on the validation set using regression performance metrics.

This validation strategy allowed the comparison of different algorithms while reducing the risk of overfitting.

7. Machine Learning Models Evaluated

The following regression algorithms were implemented and compared:

Linear Regression

Random Forest Regressor

Gradient Boosting Regressor

XGBoost Regressor

CatBoost Regressor

Each model was evaluated using regression metrics, and their predictive performance was compared.

8. Final Model Selection

Among the evaluated models, CatBoost Regressor produced the best overall performance on the validation dataset.

CatBoost was selected because it effectively handled structured tabular data and demonstrated strong predictive accuracy compared with the other regression models.

The trained CatBoost model was therefore used to generate all final predictions.

9. Prediction Generation

The selected model was used to generate predictions for:

validation.csv, producing the required validation_predictions.csv file.

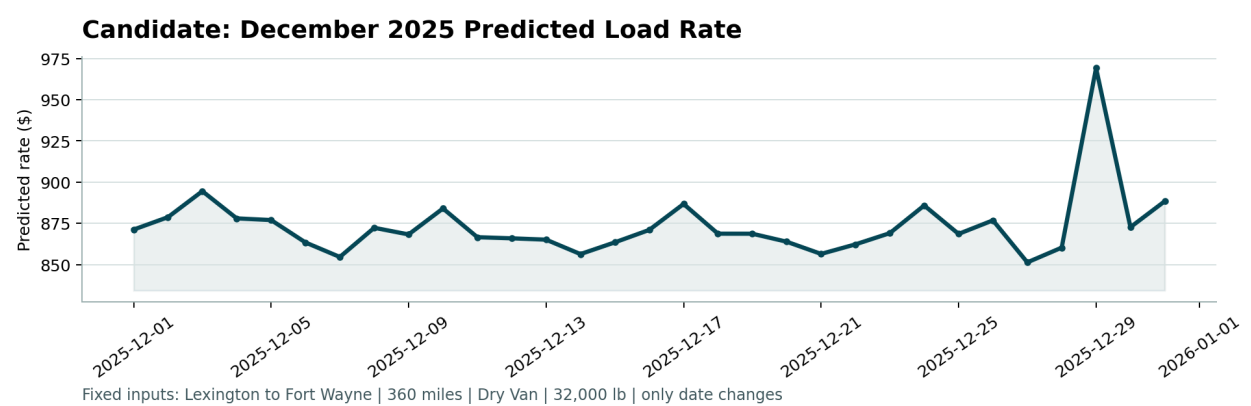
december_chart_inputs.csv, producing the December prediction file required by the scoring script.

Both prediction files were generated in the required format.

10. Model Evaluation

The provided score.py script was used to validate the prediction files.

Running the scorer successfully generated the required visualization:



candidate_december.png

This chart illustrates the predicted freight rates for the December evaluation dataset.

(Insert the generated candidate_december.png image here.)

11. Challenges and Solutions

During the project, several practical challenges were addressed:

Handling missing values within the dataset.

Engineering meaningful features from shipment dates.

Maintaining identical feature structures across multiple datasets.

Comparing several regression algorithms to identify the most suitable model.

Generating prediction files in the exact format required for submission.

12. Conclusion

This project successfully developed an end-to-end machine learning pipeline for freight rate prediction.

The workflow included data preprocessing, feature engineering, model training, validation, prediction generation, and evaluation. After comparing multiple regression algorithms, CatBoost Regressor was selected as the final model due to its superior predictive performance.

The final submission includes the prediction files, December prediction chart, source code, and documentation required for the Machine Learning Engineer assessment.

Author

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